

A Systematic Framework for Evidence-Based Color Selection

The Color Experience System™

Quantify Color. Qualify Experience.

Evidence matters. Intuition inspires.

Mark Kotterink · ORCID 0009-0001-4225-5891

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Abstract

Across architecture, healthcare, product design, and visual communication, professional color selection still rests predominantly on intuition and style, despite decades of validated research in color science, perception, and cross-cultural studies. The problem is not a shortage of evidence but the absence of a route that leads from an intended experience to a concrete, defensible color choice — the gap in which even the expert falls back on feeling.

The Color Experience System (CES) closes that gap. It is not a new color theory but the missing operational layer between research and application: a single, reproducible decision architecture that translates an intended experience into a bounded, defensible color region in CIE LCh*, operationalized in the RAL Design System+. CES proceeds in a fixed chain — concept, sense, measured baseline, cultural shift, bounded color region — in which **perception precedes meaning**: the cross-culturally stable color effect forms the baseline, while learned, culturally variable associations are carried separately and used where symbolism is precisely the point.

The framework integrates established, peer-reviewed findings rather than introducing untested claims. Its baseline rests on the colorimetric quantities that research shows to dominate color response — lightness and chroma above hue — and its cultural layer is organized through six continental clusters that shift, rather than override, that shared baseline. Every recommendation is traceable to its source, which makes CES auditable in a way that intuitive selection can never be. CES is delivered as the Atlas: an interactive working instrument, not a static lookup table.

Keywords: *color decision, evidence-based design, cross-cultural color perception, multisensory perception, color semantics, CIE LCh**

1. The Color Experience Gap

That knowledge of color carries the designer further than aptitude and experience alone is not a new claim. Green-Armytage (2006) argued that research is a more reliable route than personal judgment for knowing how people respond to color — that knowledge resolves confusion, prevents errors, and opens possibilities. Yet in professional practice, color selection still rests on feeling and intuition. This is not because designers reject knowledge, but because the knowledge does not reach them at the moment of choosing. There is no route that leads from an intended experience to a concrete color. In the absence of such a route, falling back on intuition is understandable, not a sign of incompetence.

The evidence itself is stronger than practice would suggest. Color-emotion models predict warm-cool, heavy-light, hard-soft, and active-passive directly from colorimetric quantities — the warm-cool model reaching $R^2=0.74$ — and these responses prove culture-independent across British and Chinese observers (Ou, Luo, Woodcock & Wright, 2004). Lightness (L^*) and chroma (C^*) carry the greater part of the response; hue contributes least (Valdez & Mehrabian, 1994; Gao & Xin, 2006). What is missing is not evidence but a bridge that renders it applicable. The findings are dispersed across disciplines and publications, and for even the oldest sub-question — color harmony — no generally accepted explanatory model exists (Burchett, 2002).

This gap has a visible commercial cost. At Dutch Design Week, a manufacturer observed that visitors are drawn first to color, yet ultimately buy neutral — a clean illustration of the distance between attraction and decision when no structured route connects the two. The Color Experience System is built to close that distance: to turn an intended experience into a measurable, defensible color region that a professional can stand behind.

2. What CES Is

CES is the missing operational layer between color-psychological research and applied color choice. It does not compete with colorimetric systems such as CIELAB, Munsell, or NCS, nor with reproduction standards such as Pantone; it sits above them, adding the decision layer they do not provide. CES remains fully compatible with the established colorimetric standards (CIE 15; ISO 11664-4), and expresses its output in CIE LCh* coordinates operationalized in the RAL Design System+.

The framework is the product of a cumulative, long-term synthesis of the color-science literature. Over its development, approximately 6,500 publications were collected and reviewed, from which around one hundred core studies underpin its concept-color relationships. CES organizes these validated findings into one reproducible architecture rather than introducing new theory — and that is the basis of its defensibility: every recommendation can be traced back to its source, so that a difference between two professionals reduces to a named parameter rather than to undocumented opinion.

3. A Layered Decision Architecture

A color decision does not begin with a color, but with a question about the design. In the first phase of any project, designer and client establish which effect, which affect, and which experience the design must convey. Only afterward does color enter — and in practice the order is usually reversed, color added late and guided by trends or by

whichever swatches happen to be at hand (Smith, 2003). CES intervenes in that first phase and translates the intended experience not into a single color, but into a defensible color region: a safe range within which the design may move during exploration without relinquishing its objective.

The translation proceeds in a fixed sequence:

Concept → Sense → Measured baseline → Cultural shift → Bounded LCh* color region

3.1 The entry point: concept

The chain begins with the experience itself, named as a concept — for example *vivid* or *restrained*, *calm* or *activating*. CES fixes each concept as bipolar: a positive and a negative pole, each with its own color region. A design selects not only a quality to evoke, but the side toward which it steers the experience — *vital* versus *depleted* for the life sense, *vivid* versus *sombre* for the sense of sight.

3.2 Layer one: the sensory anchor

Color is seen, but its action does not stop at sight. A color can feel warm, smell fresh, or appear calm: between color and other sensory qualities there exist stable crossmodal correspondences (Sanad, 2017), extending even to taste and flavor (Spence et al., 2010). CES uses these correspondences to anchor each concept in the sense on which it primarily acts, and to determine which color attributes carry that experience.

These sensory points of engagement are organized using Steiner's twelve-sense model, applied as a complete and non-overlapping framework of sensory domains. The empirical weight lies not in the model itself but in the crossmodal studies that, for each concept-sense pairing, substantiate the relationship between color attributes and experience. This makes CES multisensory by design — a property absent from systems built on visual parameters alone.

3.3 Layer two: the measured baseline

Anchoring a concept in a sense raises a question rather than answering it: which region of color space carries this experience? CES answers with a measured general baseline, expressed as a region in CIE LCh* and operationalized in the RAL Design System+. A concept never points to a single color but to a region; a single coordinate would imply a precision the evidence does not support.

Baseline regions are derived along disciplined routes — direct measurement where a study establishes the values, validated empirical formulae where a relationship links a color

attribute to an experience (lightness predicts the soft-hard judgment with $R^2=0.87$; Gao & Xin, 2006), and reasoned derivation from related, well-substantiated concepts where neither exists. Each region carries a designation recording how it was established, so that the basis of every recommendation is legible rather than asserted. Source studies reporting in CIELAB or Munsell are normalized by conversion, never by estimation.

3.4 Layer three: the cultural shift

The baseline describes the culture-independent structure of color experience. The cultural layer does not correct that baseline — it shifts it, describing in which direction, and with what certainty, a region deviates from the common pattern. Culture is a shift upon a shared underlying structure, not a separate region-specific system. Where no substantiated shift exists, the choice falls back on the baseline rather than on an assumption — and because the step is a shift upon a standing baseline, the magnitude of the cultural influence stays visible: the outcome records both the culture-neutral and the adjusted position.

This matters commercially as well as scientifically. A brand operating across regions can see, explicitly, where a palette must move to land correctly in a given market and where it can safely remain unchanged.

3.5 Perception precedes meaning

Within the chain, a color *effect* — the direct perceptual or bodily response that is stable across cultures — forms the baseline. An *association* — learned, symbolic, and culturally variable — weighs less and is marked separately, because it arises later in processing and is less deeply anchored. Associations are not discarded: they remain available, labeled, for applications in which symbolism is precisely what matters, such as packaging and visual communication. Unlike most of the literature, in which associations are fixed on hue alone, CES situates them across the full set of lightness, chroma, and hue. That the association-driven component is acquired rather than perceptual finds support in the ecological valence theory, in which color preference arises from the affective value of the objects a color is associated with (Palmer & Schloss, 2010).

4. Cultural Differentiation

CES employs six continental clusters as its organizing framework: EUW (Western Europe), NAM (North America), LAM (Latin America), APA (Asia-Pacific), MNA (Middle East and North Africa), and AFR (Sub-Saharan Africa). These reflect that color is not experienced identically across regions, while a shared structure dominates: across seven regions, three

factors — carried primarily by chroma and lightness — account for approximately 89% of the variance in color-emotion responses, with a very limited role for hue and cultural background (Gao et al., 2007).

The clusters therefore function not as six equivalent systems but as a structured map onto which available regional evidence is placed. Each cluster is tied to a named source, and the cultural layer is applied weighted by the reliability of that source: a shift resting on measured preference data carries more weight than one supported only by descriptive evidence, and where substantiation is absent, the baseline stands. Several clusters are defined places reserved for ongoing development, populated as the evidence base grows. This is a system that improves with use, not one that overstates what it currently knows.

5. The Framework at Work

The same chain — concept, sense, baseline, cultural shift — applies across every professional domain. Three worked illustrations show it in operation.

Built environment (EUW). A museum gallery is to feel calm and elevating, suited to contemplative viewing. The concept calm, elevating anchors in the senses of balance and sight, both pointing toward low chroma and restrained lightness rather than toward a hue. The baseline finds support in a full-scale room study in which green rooms were experienced as calm, open, and formal, the lighter low-chroma nuance judged calmer than the more chromatic one, with an identifiable elevating dimension (Hårleman, 2007); warm-white light reinforces perceived comfort and safety (Casciani & Rossi, 2013). Within the moderately chromatic Western European palette, the cultural layer barely shifts the baseline here — the outcome stays a low-chroma region, consistent with the calm intended.

Heritage (APA). The same chain, with a cluster-specific shift on the hue of warm accents. The historical “old red” of Japanese prints shifted over time toward a “new red” measured at approximately $h=17.8^\circ$, $L^*=39.3$, $C^*=64.8$ (Suzuki et al., 2013). For an Asian heritage context, a warm accent is therefore chosen not as a single fixed value, but as a baseline region shifted toward that measured position.

Packaging and product. Where symbolism and appetite drive the choice, the association layer comes forward. Color systematically shapes the perception of taste and flavor (Spence et al., 2010); CES carries these associations explicitly labeled across lightness, chroma, and hue, so that a food brand can select for a sensory and symbolic target rather than for hue alone.

The same architecture extends directly to healthcare environments, digital interfaces, and architectural specification — each a context in which an intended experience must become a concrete, defensible color, and each currently underserved by intuitive selection.

6. Why CES Is Different

	Cultural adaptation	Multisensory integration	Decision support	Relationship to CES
CIELAB	—	Visual only	Technical specification	Foundation CES builds on
Munsell	—	Visual only	Perceptual ordering	Compatible, normalized into LCh*
NCS	—	Visual only	Professional specification	Compatible reference system
Pantone	Commercial	Visual only	Reproduction	Complementary
Color Experience System	Six continental clusters, reliability-weighted	Twelve-sense organization	Concept-to-region decision chain	The decision layer above all of these

Existing systems specify color accurately; none translate an intended experience into a defensible color choice, adapt to documented cross-cultural variation, or integrate the senses beyond sight. CES adds exactly that layer — and does so on an auditable, evidence-based foundation rather than on style.

7. Output and Availability: the Atlas

CES is delivered as the Atlas — an interactive working instrument in which the chain from concept to color region is traversed directly, not a static table to be looked up. The Atlas orders colors through the L*C* diagram and the color circle, indicating for each concept its corresponding color region; from this the user selects the colors to work with. Selected colors export to the professional color systems already in workflow — NCS, RAL, RGB, and CMYK — and to PDF.

This is the form in which CES reaches practice: a tool that places the validated route between an intended experience and a concrete color directly in the hands of the professional.

8. The Color Experience Platform

The Color Experience System is the research foundation. The Color Experience Platform™, built on it, turns the chain from intended experience to bounded color region into a working tool: select the context, weight the senses, set the concepts, and read the resulting color region directly — with export to the professional color systems already in use.

CES powers evidence-based color decisions for designers, architects, brand managers, and color consultants. Request a demo and explore the Platform at **colorexperiencesystem.com**.

9. Conclusion

CES is not a new color theory but the operational layer the field has lacked: a single reproducible architecture that translates an intended experience, by way of the sense on which it acts and the cultural context in which it lands, into a bounded and defensible color region in CIE LCh*. Within that chain, perception precedes meaning, culture shifts rather than overrides a shared baseline, and every recommendation is traceable to its source. Where intuitive selection produces choices that cannot be defended or reproduced, CES produces choices that can be — and delivers them through the Atlas as a working instrument. It is presented not as a closing word but as a workable bridge between color research and color choice: built to be used, tested, and refined in professional practice.

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